Logarithm of a number

№1. Prove that $\log_2 3$ is an irrational number.

№2. Find an example of such irrational numbers a and b that a^b is a whole number.

№3. Find an example of such a rational number a and an irrational number b that a^b is a whole number.

№4. Solve:

1. $\log_2(1-x^2) = \log_2(1-x) + \log_2(1+x)$;
2. $\lg \dfrac{x^2-2x+1}{x^2+1} = \lg(x^2-2x+1) - \lg(x^2+1)$;
3. $\log_5(x^2-4x+4) = 2\log_5(2-x)$;
4. $\log_5(x^2-4x+4) = 2\log_5|x-2|$.

№5. Calculate:

1. $\lg\sin 1^\circ \cdot \lg\sin 2^\circ \cdot \lg\sin 3^\circ \cdot \ldots \cdot \lg\sin 89^\circ \cdot \lg\sin 90^\circ$;
2. $\lg\tan 10^\circ \cdot \lg\tan 15^\circ \cdot \lg\tan 20^\circ \cdot \ldots \cdot \lg\tan 75^\circ \cdot \lg\tan 80^\circ$;
3. $\lg\left(\tan 30^\circ \cdot \tan 32^\circ \cdot \tan 34^\circ \cdot \ldots \cdot \tan 58^\circ \cdot \tan 60^\circ\right)$;
4. $\lg\tan 1^\circ + \lg\tan 2^\circ + \lg\tan 3^\circ + \ldots + \lg\tan 88^\circ + \lg\tan 89^\circ$.

№6. Calculate:

1. $\log_4 5 \cdot \log_5 6 \cdot \log_6 7 \cdot \log_7 32$;
2. $\log_3 2 \cdot \log_4 3 \cdot \log_5 4 \cdot \ \ldots \cdot \log_{10} 9$.

№7. Plot the function:

1. $y = \lg\tan x + \lg\cot x$;
2. $y = x^{\log_x 2x}$;
3. $y = \log_x 1$;
4. $y = \dfrac{\lg(x^2+1)}{\lg(x^2+1)}$;
5. $y = 10^{\frac{1}{\log_x 10}}$.

№8. Plot the set of all points which satisfy the equality:

1. $\lg xy = \lg x + \lg y$;
2. $\lg xy = \lg(-x) + \lg(-y)$;
3. $\lg x^2 + \lg y^2 = 2\lg|x| + 2\lg|y|$;
4. $\log_{x^2} y^2 = \log^x(-y)$;
5. $\lg \dfrac{x}{y} = \lg(-x) - \lg(-y)$;
6. $\lg x^2 + \lg y^2 = 2\lg x + 2\lg(-y)$;
7. $\log_{x^2} y^2 = \log_x y$.

№9. Express $\log_{ab} x$ in terms of $\log_a x$ and $\log_b x$.

№10. Prove that $\dfrac{\log_a x}{\log_{ab} x} = 1 + \log_a b$.

№11. Find

1. $\log_{45} 60$ if $\log_5 2 = a$, $\log_5 3 = b$;
2. $\log_{ab} b$ if $\log_{ab} a = 4$;
3. $\log_{30} 8$ if $\lg 5 = a$, $\lg 3 = b$;
4. $\log_{60} 27$ if $\log_{60} 2 = a$, $\log_{60} 5 = b$;
5. $\log_{175} 56$ if $\log_{14} 7 = a$, $\log_5 14 = b$;
6. $\log_8 9$ if $\log_{12} 18 = a$;
7. $\log 56$ if $\lg 2 = a$, $\lg 3 = b$;
8. $\log_{150} 200$ if $\log_{20} 50 = a$, $\log_3 20 = b$.